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Question 1: DES / 3DES Encryption

Python Code:

```
from Crypto.Cipher import DES3
from Crypto.Util.Padding import pad, unpad
key = b'123456789012345678901234'
iv = b'12345678'
cipher = DES3.new(key, DES3.MODE_CBC, iv)
message = b"Hello World"
ciphertext = cipher.encrypt(pad(message, 8))
print(ciphertext.hex())
cipher = DES3.new(key, DES3.MODE_CBC, iv)
print(unpad(cipher.decrypt(ciphertext),8).decode())
```

Encrypted: 3963b1ec787dc0b5bbbed2c0372c535e4

Decrypted: Hello World

Question 2: AES CBC Mode

Python Code:

```
from Crypto.Cipher import AES
from Crypto.Util.Padding import pad, unpad
key=b'1234567890123456'
iv=b'abcdefghijklmnp'
cipher=AES.new(key,AES.MODE_CBC,iv)
ct=cipher.encrypt(pad(b'Secure Message',16))
print(ct.hex())
```

```
cipher=AES.new(key,AES.MODE_CBC,iv)
print(unpad(cipher.decrypt(ct),16).decode())
```

Encrypted: 80c9f7ec94c5165d1497cd49b00be55b

Decrypted: Secure Message

Question 3: RSA Encryption

Python Code:

```
from Crypto.PublicKey import RSA
from Crypto.Cipher import PKCS1_OAEP
```

```
key=RSA.generate(2048)
cipher=PKCS1_OAEP.new(key.publickey())
ct=cipher.encrypt(b'Hello RSA')
print(ct.hex())
print(PKCS1_OAEP.new(key).decrypt(ct).decode())
```

```
Encrypted:
57a2eb2495946c669acc14413f3d4a5186fb98fb5a9fc68e9b7a16862de4fc69befcfb0468ef8ae
2f338a42f6622262ce571432eadd14ce4baede7d3a7ff2b4ed35e7316846fab250bec54dfcc7ccf
ec5b353d0d3041cc7b336f4ff3327f8d598223ddadd07c34e43624d60b4787fccc5d5ad46512526
dbf90a38fa68a7f7d96944ebcf693a6c31e25ebdceca7c06c3cb2a051493302cd1f5f8c93d5352
0137ef4e9384959a7b2031ec4de3d802d051a993fba9a41363726a5a96fdcedc16757c278f9b3e4
cc273bf372be1376530ef182a40ef94151e29ca0263bbab3607af6b525f7eecf5a37483c89c6dd1
5f6123d857c73651567433458b7149a6870754
Decrypted: Hello RSA
```

Question 4: Diffie-Hellman

Python Code:

```
p=23
g=5
a=6
b=15
A=pow(g,a,p)
B=pow(g,b,p)
```

```
print(pow(B,a,p))
print(pow(A,b,p))
```

```
Alice Public Key: 8
Bob Public Key: 19
Alice Shared Key: 2
Bob Shared Key: 2
Key Exchange Successful
```

Question 5: ECC

Python Code:

```
from tinyec import registry
import secrets
```

```
curve=registry.get_curve('brainpoolP256r1')
a=secrets.randbelow(curve.field.n)
b=secrets.randbelow(curve.field.n)
A=a*curve.g
B=b*curve.g
print((a*B).x)
print((b*A).x)
```

```
Alice Shared Key:
50857714759168544848602594980813021021498109713322151740672471902270544425917

Bob Shared Key:
50857714759168544848602594980813021021498109713322151740672471902270544425917

Key Exchange Successful
```